Lab 4 practice: undetermined coefficients (Zill §3.4)

Prep for the Wed Oct 7 lab quiz. Closed book, about 10 minutes per problem. Part A (problems 1 to 8): solve. Part B: write the form of y_p only, or say the method does not apply and why. Every solution was checked by substituting back (`lab4_check.py`).

Homogeneous part. For $ay'' + by' + cy = 0$ solve $am^2 + bm + c = 0$:

Roots	y_c
distinct real m_1, m_2	$c_1 e^{m_1 x} + c_2 e^{m_2 x}$
repeated real m	$(c_1 + c_2 x)e^{mx}$
complex $\alpha \pm \beta i$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

Guess for y_p.

$g(x)$	y_p
constant	A
polynomial of degree n	$A_n x^n + \cdots + A_1 x + A_0$ (every power)
e^{ax}	Ae^{ax}
$\cos \beta x$ or $\sin \beta x$	$A \cos \beta x + B \sin \beta x$ (always both)
product of the above	product of the guesses
sum of the above	solve each piece separately, then add

Overlap rule. If any term of the guess also appears in y_c, multiply the whole guess by x^s, where s is the smallest integer that removes every overlap (the multiplicity of the matching root).

Finish. $y = y_c + y_p$. Apply initial conditions last, to the whole y.

Problem 1

$y'' + 3y' + 2y = 6x^2 - 3$

Solution: polynomial forcing, no overlap

Second order, linear, constant coefficients. $m^2 + 3m + 2 = (m+1)(m+2) = 0$, so

$$y_c = c_1 e^{-x} + c_2 e^{-2x}$$

Try the full quadratic $y_p = Ax^2 + Bx + C$. Zero is not a root, so there is no overlap. Then $y_p' = 2Ax + B$ and $y_p'' = 2A$:

$$2A + 3(2Ax + B) + 2(Ax^2 + Bx + C) = 2Ax^2 + (6A + 2B)x + (2A + 3B + 2C)$$

Match against $6x^2 + 0x - 3$:

$$2A = 6, \qquad 6A + 2B = 0, \qquad 2A + 3B + 2C = -3$$

So $A = 3$, $B = -9$, and $6 - 27 + 2C = -3$ gives $C = 9$.

$$\boxed{y = c_1 e^{-x} + c_2 e^{-2x} + 3x^2 - 9x + 9, \qquad -\infty < x < \infty}$$

The Bx term is needed even though g has no x term. Leave it out and the x terms on the left have nothing to balance them.

Problem 2

$y'' - 2y' - 3y = 6e^{2x}$

Solution: exponential forcing, no overlap

$m^2 - 2m - 3 = (m-3)(m+1) = 0$, so $y_c = c_1 e^{3x} + c_2 e^{-x}$.

The forcing is e^{2x} and 2 is not a root, so $y_p = Ae^{2x}$ with no modification:

$$(4A - 4A - 3A)e^{2x} = -3Ae^{2x} = 6e^{2x} \quad \Longrightarrow \quad A = -2$$

$$\boxed{y = c_1 e^{3x} + c_2 e^{-x} - 2e^{2x}}$$

Problem 3

$y'' - y' - 2y = 10\cos x$

Solution: sine and cosine together

$m^2 - m - 2 = (m-2)(m+1) = 0$, so $y_c = c_1 e^{2x} + c_2 e^{-x}$.

The forcing is only a cosine, but the y' term turns cosine into sine, so the guess needs both: $y_p = A\cos x + B\sin x$. The roots are real, so $\pm i$ is not a root and there is no overlap.

$$y_p' = -A\sin x + B\cos x, \qquad y_p'' = -A\cos x - B\sin x$$

Collect the $\cos x$ and $\sin x$ terms of $y_p'' - y_p' - 2y_p$:

$$\cos x: \ -A - B - 2A = -3A - B = 10 \qquad\qquad \sin x: \ -B + A - 2B = A - 3B = 0$$

The second gives $A = 3B$, then $-10B = 10$, so $B = -1$ and $A = -3$.

$$\boxed{y = c_1 e^{2x} + c_2 e^{-x} - 3\cos x - \sin x}$$

Problem 4

$y'' - 3y' + 2y = 4e^x$

Solution: resonance with a simple root

$m^2 - 3m + 2 = (m-1)(m-2) = 0$, so $y_c = c_1 e^x + c_2 e^{2x}$.

The forcing e^x is already a term of y_c ($m = 1$ is a simple root). The naive guess Ae^x would make the left side zero, so multiply by x: $y_p = Axe^x$.

$$y_p' = A(x+1)e^x, \qquad y_p'' = A(x+2)e^x$$

$$Ae^x\big[(x+2) - 3(x+1) + 2x\big] = -Ae^x = 4e^x \quad \Longrightarrow \quad A = -4$$

$$\boxed{y = c_1 e^x + c_2 e^{2x} - 4xe^x}$$

Problem 5

$y'' - 6y' + 9y = 4e^{3x}$

Solution: repeated root, so x^2

$m^2 - 6m + 9 = (m-3)^2 = 0$, so $y_c = (c_1 + c_2 x)e^{3x}$.

Now *both* e^{3x} and xe^{3x} are in y_c. Multiplying the guess once by x gives Axe^{3x}, which is still a term of y_c. It has to be x^2: $y_p = Ax^2 e^{3x}$.

$$y_p' = A(2x + 3x^2)e^{3x}, \qquad y_p'' = A(2 + 12x + 9x^2)e^{3x}$$

$$Ae^{3x}\big[(2 + 12x + 9x^2) - 6(2x + 3x^2) + 9x^2\big] = 2Ae^{3x} = 4e^{3x} \quad \Longrightarrow \quad A = 2$$

$$\boxed{y = (c_1 + c_2 x)e^{3x} + 2x^2 e^{3x}}$$

Problem 6

$y'' + 9y = 12\cos 3x$

Solution: complex roots, forcing at the natural frequency

$m^2 + 9 = 0$ gives $m = \pm 3i$, so $y_c = c_1 \cos 3x + c_2 \sin 3x$.

The forcing $\cos 3x$ and the guess $A\cos 3x + B\sin 3x$ both sit inside y_c, so the whole pair gets an x: $y_p = x(A\cos 3x + B\sin 3x)$.

$$y_p'' = 2(-3A\sin 3x + 3B\cos 3x) + x(-9A\cos 3x - 9B\sin 3x)$$

The terms with a factor of x cancel against $9y_p$, leaving

$$y_p'' + 9y_p = -6A\sin 3x + 6B\cos 3x = 12\cos 3x \quad \Longrightarrow \quad A = 0,\ B = 2$$

$$\boxed{y = c_1 \cos 3x + c_2 \sin 3x + 2x \sin 3x}$$

The $2x\sin 3x$ term grows without bound. That is resonance, and it is why the x factor is unavoidable.

Problem 7

$y'' - 4y' + 3y = x + xe^x$

Solution: superposition, with one piece overlapping

$m^2 - 4m + 3 = (m-1)(m-3) = 0$, so $y_c = c_1 e^x + c_2 e^{3x}$.

The forcing is two unlike pieces, so solve for each separately: $g_1 = x$ and $g_2 = xe^x$.

For $g_1 = x$. Zero is not a root, so use $Cx + D$:

$$0 - 4C + 3(Cx + D) = 3Cx + (3D - 4C) = x \quad \Longrightarrow \quad C = \tfrac{1}{3},\ D = \tfrac{4}{9}$$

For $g_2 = xe^x$. The natural guess is $(Ax + B)e^x$, but e^x is in y_c, so multiply by x: $y_{p2} = x(Ax + B)e^x$. Write it as ue^x with $u = Ax^2 + Bx$. Then $y' = (u + u')e^x$ and $y'' = (u + 2u' + u'')e^x$, and

$$y'' - 4y' + 3y = (u'' - 2u')e^x$$

Set $u'' - 2u' = x$, which is $2A - 4Ax - 2B = x$. So $A = -\frac{1}{4}$, and $2A - 2B = 0$ gives $B = -\frac{1}{4}$.

Add the pieces:

$$\boxed{y = c_1 e^x + c_2 e^{3x} + \frac{x}{3} + \frac{4}{9} - \frac{x^2 + x}{4} e^x}$$

Problem 8

$y'' + 4y' + 5y = 35e^{-4x}, \qquad y(0) = -3, \quad y'(0) = 1$

Solution: initial value problem with complex roots

$m^2 + 4m + 5 = 0$ gives $m = -2 \pm i$, so $y_c = e^{-2x}(c_1 \cos x + c_2 \sin x)$.

The forcing e^{-4x} does not appear in y_c, so $y_p = Ae^{-4x}$:

$$(16A - 16A + 5A)e^{-4x} = 5Ae^{-4x} = 35e^{-4x} \quad \Longrightarrow \quad A = 7$$

General solution first, then the conditions:

$$y = e^{-2x}(c_1 \cos x + c_2 \sin x) + 7e^{-4x}$$

$y(0) = c_1 + 7 = -3$, so $c_1 = -10$. Differentiating,

$$y' = e^{-2x}\big[(-2c_1 + c_2)\cos x + (-2c_2 - c_1)\sin x\big] - 28e^{-4x}$$

$y'(0) = -2c_1 + c_2 - 28 = 20 + c_2 - 28 = 1$, so $c_2 = 9$.

$$\boxed{y = e^{-2x}(9 \sin x - 10 \cos x) + 7e^{-4x}, \qquad -\infty < x < \infty}$$

Applying $y(0) = -3$ to y_c alone would give $c_1 = -3$, which is the classic wrong answer here.

Part B (a)

$y'' + 4y = x \sin 2x$

Solution

Roots $\pm 2i$, so $y_c = c_1 \cos 2x + c_2 \sin 2x$. The base guess for $x \sin 2x$ is $(Ax + B)\cos 2x + (Cx + D)\sin 2x$, which contains $B\cos 2x$ and $D\sin 2x$, both in y_c. Multiply the whole thing by x:

$$y_p = x\big[(Ax + B)\cos 2x + (Cx + D)\sin 2x\big]$$

Part B (b)

$y''' - y'' = 4x + e^x$

Solution

$m^2(m-1) = 0$, so the roots are $0, 0, 1$ and $y_c = c_1 + c_2 x + c_3 e^x$. Piece by piece:

- $4x$: the base guess $Ax + B$ collides with $c_1 + c_2 x$. Zero is a double root, so $s = 2$ and you get $x^2(Ax + B)$.
- e^x: 1 is a simple root, so Cxe^x.

$$y_p = x^2(Ax + B) + Cxe^x$$

Part B (c)

$y'' + y = \sec x$

Solution

The method does not apply. $\sec x$ is not a polynomial, exponential, sine, cosine or a product of them, and its derivatives never settle into a finite family to guess from. Use variation of parameters (Lab 5, §3.5).

Part B (d)

$y'' + 2y' + 5y = e^{-x} \sin 2x$

Solution

Roots $-1 \pm 2i$, so $y_c = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$. The base guess $e^{-x}(A \cos 2x + B \sin 2x)$ is exactly the form of y_c, so multiply by x:

$$y_p = xe^{-x}(A \cos 2x + B \sin 2x)$$

Part B (e)

$x^2y'' + xy' - y = 3x$

Solution

The method does not apply as taught. The coefficients are functions of x, not constants, and the table of guesses assumes constant coefficients.